Listing 3-1. *7-Color Blink*

```
const int red = 9;
const int green = 10;
const int blue = 11;

int time = 1000;
int multiple = 2;

void setup() {
  pinMode(red, OUTPUT);
  pinMode(green, OUTPUT);
  pinMode(blue, OUTPUT);
}

void loop() {
  digitalWrite(red, HIGH);     // red
  delay(time);
  digitalWrite(green, HIGH);   // yellow
  delay(time);
  digitalWrite(red, LOW);      // green
  delay(time);
  digitalWrite(blue, HIGH);    // cyan
  delay(time);
  digitalWrite(green, LOW);    // blue
  delay(time);
  digitalWrite(red, HIGH);     // magenta
  delay(time);
  digitalWrite(green, HIGH);   // white
  delay(time*=multiple);
  digitalWrite(blue, LOW);     // reset
  digitalWrite(green, LOW);
  time/=multiple;
}
```

Source Code Summary

To kick things off, we start our sketch with a block of code that defines each of the variables that will be used in our code. In this way we will know, for example, that the red LED is connected to pin 9 and the time delay between colors is 1000 milliseconds, or 1 second. If we wanted to change any of these values, to set different pin numbers for the RGB LED or to increase or decrease the delay time, we would only have to change the value in one single location, speeding up changes to the sketch and making it that much easier to understand. This will become more important later on when our sketches begin to do more interesting things with our variables, but for now let's look at the following first line of code in more detail:

```
const int red = 9;
```